

Illuminating Pneumonia: Explainable TinyML-based Pneumonia Diagnosis Framework for Digital Health

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Abstract— To address the ever-growing need for automation in the healthcare sector, this paper presents a robust deep learning-based framework utilizing TinyML for deployment in resource-constrained environments. Building upon the concept of transparency, we introduced Explainable AI (XAI) to ensure that the model doesn't just predict like a 'black-box' but also explains why it made that prediction. We trained various models and evaluated them using different metrics to select the best model after quantization. Within this framework, the XAI techniques of Local Interpretable Model-Agnostic Explanations (LIME), Shapley Value Sampling, and Occlusion were integrated. The proposed framework provides a balanced approach between diagnostic accuracy and computational efficiency. Specifically, the quantized version of MobileNetV3 showed a strong balance between accuracy and on-device deployment. It achieved 98.24% accuracy and an inference time of 1.417 milliseconds, making it the best architecture for the framework.

Keywords— *Pneumonia Detection, Explainable AI, Quantization, Healthcare AI, On-device AI, Image Classification*

I. INTRODUCTION

Attention towards healthcare has always been of utmost importance. Especially in today's world, our lifestyle has played a significant role in the decline of fitness and productivity. Such concerns lead up to the health revolutions from Health 1.0 to Health 5.0. Technological reforms have furthermore reformed this evolution and significantly transformed medical diagnosis, treatment, and patient care. Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have led to more optimized opportunities in reduced subjectivity in diagnosis, faster results, and accuracy beyond human limits.

Although this wave of evolution has revolutionized the detection and treatment of many chronic diseases, as well as those characterized by diagnostic complexity, pneumonia remains a global health challenge. Pneumonia is an inflammatory condition of the lungs, caused by bacterial,

viral, or fungal pathogens. It is a severe lung infection that causes inflammation and fluid buildup in the alveoli, which is

visible on chest X-rays. If untreated, pneumonia can lead to life-threatening complications. Estimates of global deaths due to pneumonia are around 2.18 million, with an increase to 2.5 million during the COVID-19 pandemic. In such situations, traditional methods of X-rays or Gram staining cause high rejection of samples, lower productivity, and ultimately delayed results and treatment. To overcome such scenarios, ML and DL techniques were widely utilized for pneumonia detection and classification. Previous research has used machine learning to highlight the potential of medical imaging using supervised and unsupervised methods like regression, Support Vector Machines (SVM), random forests, Convolutional Neural Networks (CNN), clustering, and dimensionality reduction. Another research by Aradhya et al. [1] used a cluster-based approach of chest X-ray images with limited training data. The method leverages the fast learning and generalization capabilities of conventional networks like Generalized Regression Neural Network (GRNN) and Probabilistic Neural Network (PNN), combined in an ensemble.

Even DL has promised significant improvement, as Islam et al. [2] have investigated using deep neural networks along with classification methods and achieved an accuracy of 98.99% using SqueezeNet + InceptionV3. Some others focused on Vision Transformers (ViT) introduced by Dosovitskiy et al. [3] to enable accurate statistics or improvement in classification of Pneumonia. Framework proposed by Ha et al. [4] achieved accuracy of 94% using ViT.

Kim et al. [5] showcased that DL algorithms aided in pneumonia detection at a higher Area under the Receiver Operating Characteristic Curve (AUROC) of 0.861 compared to physicians, achieving an AUROC of 0.788. They further showed that when they work symbiotically, they achieve way better accuracy.

In spite of such great achievements, they posed some dissatisfaction in terms of large datasets requiring longer training times and poor dataset quality, resulting in delayed results. DL algorithms lack transparency and behave like a black box.

To counter this issue, Explainable AI (XAI) was introduced to ensure transparency in prediction by understanding the factors that led the model to make the prediction. Along with XAI, we also introduce the application of TinyML to build ML models to reduce computation requirements and latency of the model to function. TinyML compressed model size in TensorFlow since DL based classification is a resource- intensive task.

A. Research Contributions

The main contributions in this work are:

- We propose the framework using the MobileNetV3 model architecture for automation of Pneumonia Classification. By using it, we are able to make a lightweight yet powerful model for deployment in real-world clinical settings where computational resources may be limited.
- We applied post-training quantization to adapt the model for TinyML applications, ensuring efficient operation on resource-constrained embedded systems while preserving high diagnostic accuracy.
- We integrated XAI to enhance transparency and interpretability in our DL model. Moving beyond "black box" predictions, we offer a clear, rational explanation for the predictions that can be scrutinized and validated by medical professionals.

B. Paper Organization

Section II is our proposed framework, which includes the explanation and pipeline structure of the classification model and application of XAI at the time of inference. The experimental setup and the performance analysis are included in Section III. Section IV is the conclusion and future work of the paper.

II. PROPOSED FRAMEWORK

Fig. 1 visualized the proposed TinyML and XAI-based framework for Pneumonia Classification. The proposed framework is separated into different parts, each for a specific task. The framework starts from the data acquisition layer to the inference and XAI layer. The detailed descriptions of each layer are provided in the following subsections.

A. Data Acquisition Layer

At the beginning, we acquired and preprocessed X-ray images to train the model. The dataset was acquired from the Chest X-Ray Images (Pneumonia) dataset from Kaggle. The dataset was made up of 1,070 normal images and 3,116 pneumonia images. It was further divided into 3 parts of 80% for training the model and 10% each for validation and testing of the model.

After data acquisition, the data is preprocessed to ensure that the model is trained as expected. It involves the following steps:

- Converting to RGB: The images are converted to RGB pixels for model training. A grayscale image is converted to RGB form in the following manner [6].

Let $I_G(x, y)$ be the intensity of a pixel at coordinates (x, y) in the grayscale image. The corresponding RGB pixel, $I_{RGB}(x, y)$,

is defined by setting its R, G, and B components equal to the grayscale intensity.

$$I_{RGB}(x, y) = (R, G, B): \begin{cases} R(x, y) = I_G(x, y) \\ G(x, y) = I_G(x, y) \\ B(x, y) = I_G(x, y) \end{cases} \quad (1)$$

- Rescaling Images The original images are rescaled to a standard size so that they fit into the model. The image rescaling occurs as follows [6].

Let the original image, I_{orig} , is of dimensions $W_{orig} \times H_{orig}$. The new image, I_{new} , will be of dimensions $W_{new} \times H_{new} = 128 \times 128$. For each pixel (x', y') in the new image, where $0 \leq x' < W_{new}$ and $0 \leq y' < H_{new}$, its corresponding coordinates (x, y) in the original image are calculated using the scaling factors:

$$x = x' \cdot \frac{W_{orig}}{W_{new}} \text{ and } y = y' \cdot \frac{H_{orig}}{H_{new}} \quad (2)$$

$$x = x' \cdot \frac{W_{orig}}{128} \text{ and } y = y' \cdot \frac{H_{orig}}{128} \quad (3)$$

- Pixel Scaling: The pixels are scaled down into the range of 0 to 1 from the range of 0 to 255. This process is done as follows [6].

We represent the original and scaled pixels by the vectors $p_{original}$ and p_{scaled} , defined as (4) and (5):

$$\overrightarrow{p_{original}} = (R_{original}, G_{original}, B_{original}) \quad (4)$$

$$\overrightarrow{p_{scaled}} = (R_{scaled}, G_{scaled}, B_{scaled}) \quad (5)$$

Then, we can obtain the vector p_{scaled} from $p_{original}$ by the (6):

$$\overrightarrow{p_{scaled}} = \frac{\overrightarrow{p_{original}}}{255.0} \quad (6)$$

B. Training and Quantization

After the data are acquired and preprocessed, the training and quantization process of the model starts. To ensure that we choose the best model architecture, we selected MobileNetV3 [7], SimpleCNN, EfficientNetB0 [8], ResNet50 [9], DenseNet121 [10], Visual Geometry Group - 16 layers (VGG16) [11]. After training, the model that gave the best performance after quantization will be selected.

1) *MobileNetV3*: The MobileNetV3 architecture was proposed by Andrew et al. [7] for mobile and embedded vision applications. It is based on depthwise separable convolutions, a form of factorized convolution. In this architecture, all layers are followed by Rectified Linear Unit (ReLU) [12] nonlinearity, except in the last fully connected layer, which feeds directly into a softmax layer for classification.

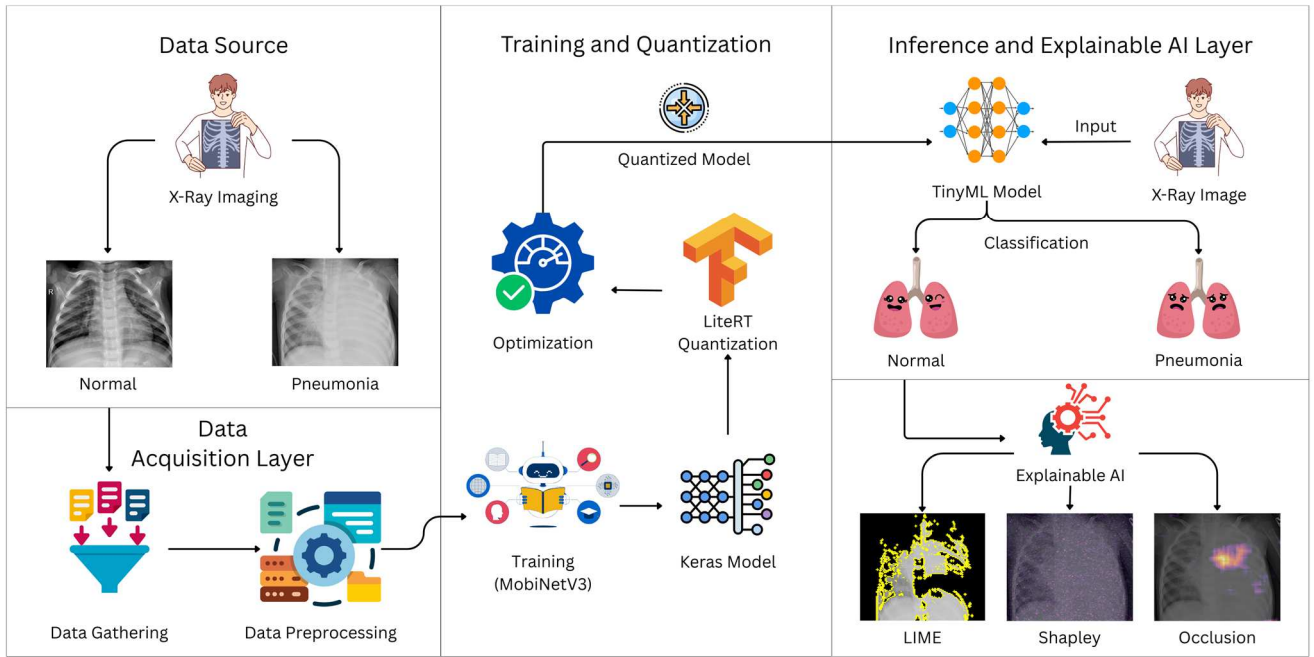


Fig. 1. Proposed framework

$$ReLU(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (7)$$

$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} \quad (8)$$

For model training, we used the Adam optimizer with a fixed learning rate (η) of 1×10^{-4} . We have also applied Binary Cross-Entropy (BCE) [13] loss to quantify the difference between the true class labels (0 or 1) and the model's predicted probabilities.

The BCE loss is defined as (9):

$$BCE = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (9)$$

We utilized this model architecture after comparing it with other architectures and found it best suited for embedded vision applications after rigorous performance testing of the models.

2) *Quantization*: Quantization [14] is a technique utilized in deep learning models to convert weights and activation values from high precision to a lower precision. Usually a 32-bit floating point (FP32) or a 16-bit floating point (FP16), to an 8-bit integer (INT8). By going from a high-precision to a lower precision, the data is compressed to a lower precision level, resulting in lower accuracy. This loss is known as quantization error.

The quantization of floating-point tensors to lower precision is done using (10):

$$Q(x, scale, zero_point) = \text{round} \left(\frac{x}{scale} - zero_point \right) \quad (10)$$

Here, x is the floating point full-precision value of the tensor, $scale$ is the conversion factor, and $zero_point$ is the value representing 0.0 in full precision. The $Q(x, scale, zero_point)$ is the final quantized tensor derived from x . By quantization, the footprint of the model is reduced, leading to faster inference, more efficiency, and lower power consumption. We performed post-training quantization, where the weights and activations are quantized ahead of time. The lower precision point was chosen using the LiteRT Framework for the optimum quantized model.

C. Inference and Explainable AI layer

The final quantized model is deployed for the classification of pneumonia. Being a binary classification model, the model will give output as 0 for normal or 1 for pneumonia for the X-ray image given to it as input.

$$y = \begin{cases} Normal & \text{if } y = 0 \\ Pneumonia & \text{if } y = 1 \end{cases} \quad (11)$$

While utilization of DL models is already in use, we introduced XAI on quantized versions of the models to ensure transparency in model prediction. By implementing XAI on models, we ensured that the model not only predicts, but also provides an explanation about the factors that lead to the prediction. We applied the following three XAI algorithms:

- **LIME**: LIME [15] is an explanation technique that explains the predictions of any classifier in an interpretable and faithful manner, by learning an interpretable model locally around the prediction.
- **Occlusion**: Occlusion [16] is a perturbation-based explanation approach to compute attribution, done by replacing each contiguous rectangular region with a

given baseline to it, and computing the difference in the model's prediction.

- **Shapley Value Sampling:** Shapley Value Sampling [17] is an attribution method based on concepts from cooperative game theory. This technique involves permutations of input features and adding them sequentially to a given baseline. The variation in prediction after adding each one corresponds to its contribution to the prediction.

III. RESULTS AND DISCUSSION

A. Experimental Setup

This experiment was conducted on the Kaggle platform, utilizing a virtual environment with the following specifications: Intel Xeon Processor @ 2.00GHz, 30 GB RAM, and NVIDIA T4 16GB VRAM GPU. The complete implementation was carried out in the Python environment v3.11.13, using the following libraries: TensorFlow (v2.18.0) [18], NumPy (v1.26.4), Pandas (v2.2.3), Scikit-learn (v1.2.2), Matplotlib (v3.7.2), OpenCV (v4.11.0), Scikit-image (v0.25.2), psutil (v7.0.0), LIME (v0.2.0.1), and tqdm (v4.67.1).

B. Performance Analysis

The proposed framework and the component selection were done based on performance metrics of the models for the task. The experiment performance analysis was done at every single step, from training the model to quantization and explanations by XAI during inference. The results are described below.

1) **Model Training Results:** Fig. 2 shows the training and validation accuracy and loss over epochs to visualize the model training results. The DenseNet121 model achieved the best accuracy while training.

2) **Quantized Model Comparison with Normal Model:** Fig. 3a shows the accuracy comparison of Normal models with their corresponding quantized versions of the models. We can see that the DenseNet121's quantized version was the best performer, followed by MobileNetV3 and ResNet50.

Fig. 3c shows the decrease in RAM required between normal and quantized models. The DenseNet121's quantized

version showed the best results in optimizing RAM requirements.

Fig. 3d shows the storage space acquired by the normal models and their quantized versions. Here, the SimpleCNN, being a primitive model, performed the best, followed by MobileNetV3.

Fig. 3b showcases the inference time of the models. The MobileNetV3's quantized version outperforms other quantized models.

3) **Quantized Model Comparison with Normal Model:** Quantization is performed to balance the model's performance while keeping environmental constraints in mind. Thus, we compared quantized models with their original versions for

their accuracy, storage space, inference time, and RAM usage. The results of this comparison are as discussed.

Fig. 4 Shows the model performance of the quantized versions of the models. The DenseNet121's quantized model achieved the best performance. Being quite large in storage size, it is rejected as the proposed framework is built targeting a constrained environment.

Fig. 5 shows the comparison of the quantized model's performance on the following metrics: accuracy, inference time, model storage size, and RAM usage.

4) **XAI Results:** By using XAI, we validate the model's diagnostic prediction by understanding which portion led to the prediction it made. We employed three post-hoc following XAI techniques: LIME, Shapley Value Sampling, and Occlusion. These methods produce explanation heatmaps highlighting the regions most influential to the model's classification prediction.

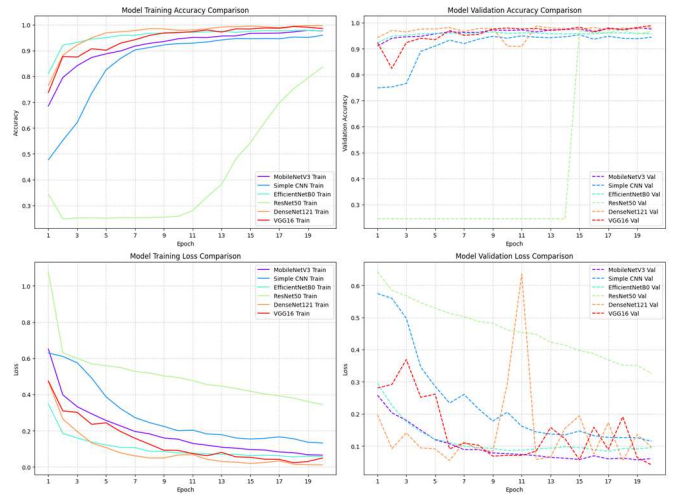


Fig. 2. Top-left: Training accuracy over epochs of models during training, Top-right: Validation accuracy over epochs of models during training, Bottom-left: Training loss over epochs of models during training, Bottom-right: Validation loss over epochs of models during training

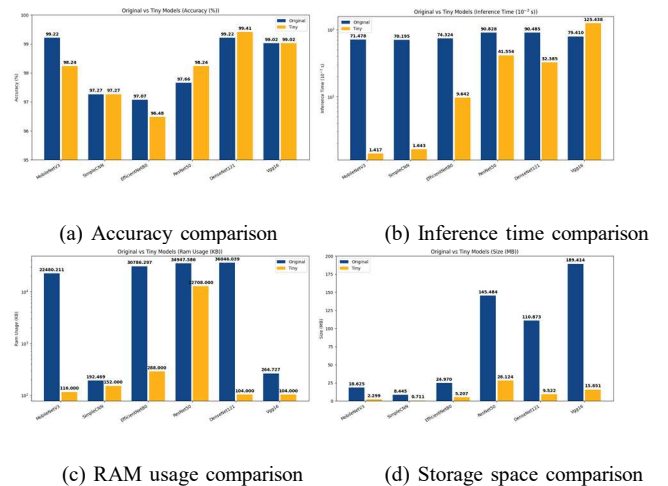


Fig. 3. Performance comparison between normal and quantized models. Top-left: Accuracy, Top-right: Inference time, Bottom-left: RAM, Bottom-right: Storage.

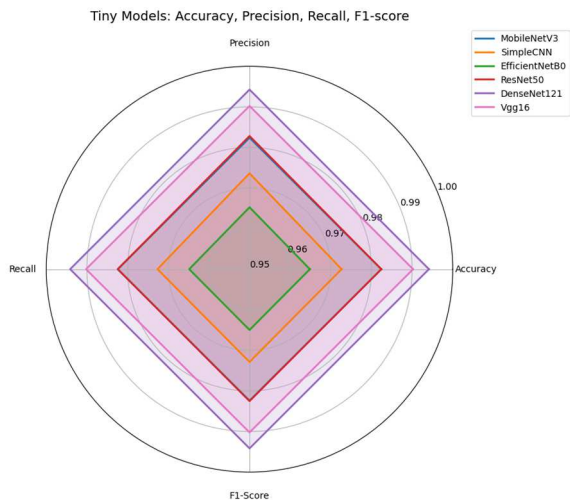


Fig. 4. Radar plot visualizing performance of quantized models in Accuracy, F1-Score, Recall, and Precision

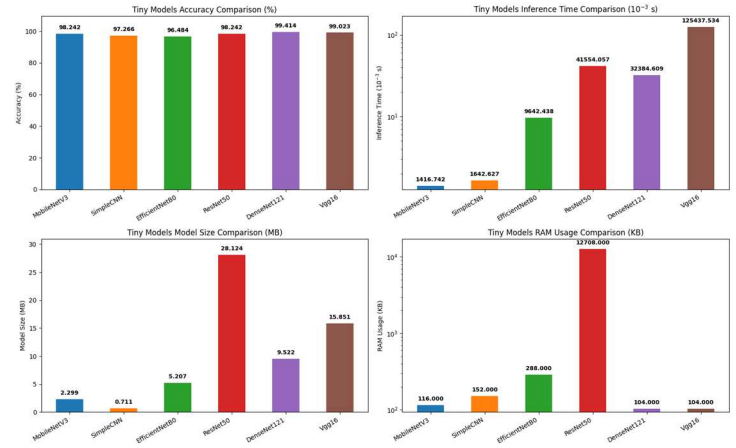


Fig. 5. Top-left: Bar graph of accuracy of quantized models, Top-right: Bar graph of inference time of quantized models, Bottom-left: Bar graph of storage space required for quantized models, Bottom-right: Bar graph of RAM used by quantized models

TABLE I. EXPLANATION HEATMAP OVERLAY ON X-RAY IMAGES FOR LIME, SHAPLEY VALUE SAMPLING, AND OCCLUSION ALGORITHMS FOR NORMAL, BACTERIAL PNEUMONIA, AND VIRAL PNEUMONIA

	LIME	Shapley Value Sampling	Occlusion
Normal			
Bacterial Pneumonia			
Viral Pneumonia			

The primary goal of this analysis is to determine whether the models focus on the actual pathological indicators visible in the radiographs. Table I shows the explanations provided by LIME, Shapley Value Sampling, and Occlusion for a sample image with Normal, Bacterial Pneumonia, and Viral Pneumonia. The results are explained as follows:

- For the Normal X-ray images, LIME highlighted that the lungs are proper, whereas the Shapley Value Sampling and Occlusion produced a diffused highlight. This is the expected behavior, as there is no specific region of pathology for the model to focus on.

- For the Pneumonia X-ray images, LIME highlighted the light parts of the image, and the Occlusion gave more highlight intensity in the lighter portion. Here, the Shapley Value Sampling algorithm wasn't able to perform at par. This is explained. In X-ray, the air-filled lungs allow X-ray images to pass, whereas the pneumonia-affected lungs are hazy white because the fluid-filled lungs reflect the X-rays.

IV. CONCLUSION

In this paper, we propose the application of a quantized deep learning model for Pneumonia detection using X-ray images. Along with model quantization, we proposed the application of XAI to ensure transparency in model prediction. The framework utilizes LiteRT for the quantization of a Keras model trained on a dataset and applies LIME for XAI in the model. This way, our proposed framework is both robust and transparent to answer the current needs in the medical industry. The results and comparison of model training, quantization, and application are shown in the Results and Discussion section. While the proposed study rigorously evaluates CNN architectures, like ViTs within the TinyML pipeline. The current dataset is limited to 4,186 images, necessitating further research using more extensive and diverse clinical data to ensure broader generalizability. To improve the framework's robustness, future work will focus on scaling the dataset and implementing continual learning to ensure the model adapts with evolving medical applications. Furthermore, the scope will be expanded to incorporate transformer-based architectures and multi-disease capabilities to enhance automated clinical workflows in Healthcare 5.0.

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